TECHNICAL STUDY GUIDE

The Field Guide to Global Payments

Complete Visual Breakdown of Core Rails, Economics, Risk Architecture & Business Models

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SECTION 01

How Money Moves in a Card Transaction (The Four-Party Model)

1. Cardholder
Initiates purchase with credit or debit card
Payer

2. Merchant
Presents terminal or checkout; submits auth payload
Payee

3. Acquirer / Processor
Underwrites merchant; formats ISO 8583 message to network
Merchant Bank

4. Card Network
Visa / Mastercard / Amex: switches & clears transaction
The Rails

5. Issuing Bank
Holds cardholder funds; approves/declines; guarantees payout
Consumer Bank

Auth Request → Acquirer Switch → Network Switch → Issuer Auth Approval →

Phase 1 Authorization (Seconds)

- **Auth Hold**: Issuer confirms funds/credit line and places temporary hold.
- **Real-time Checks**: CVV match, AVS (Address Verification), 3D-Secure challenge.
- **Decline Types**: Soft declines (insufficient funds, retryable) vs Hard declines (stolen card, closed account).

Phase 2 Clearing & Settlement

- **Batch Capture**: Merchant batches approved auths at end-of-day.
- **Gross-to-Net**: Network routes clearing messages, nets balances between banks.
- **Payout**: Acquirer deducts fees and pays net funds into merchant bank account (T+1 to T+3).

Fee Waterfall Economic Cut

- **Interchange (~1.5%–2.5%)**: Paid directly to Issuer for credit risk & loyalty rewards.
- **Assessment Fee (~0.13%–0.15%)**: Paid to Visa/Mastercard for rail usage.
- **Acquirer Markup (e.g. 0.10% + $0.10)**: Paid to processor for gateway & servicing.

Liability & Disputes: Acquirer assumes credit risk if merchant goes bankrupt before fulfillment. Under 3D Secure (EMV 3DS), fraud liability shifts from the merchant to the card issuer.

SECTION 02

Non-Card Payment Method Taxonomy

A2A / Bank Rails

Direct bank transfers bypassing card rails. Lower fee, high consumer trust.

- **Instant Push**: UPI (India), Pix (Brazil), Faster Payments (UK). Instant finality.
- **Direct Debit / Pull**: ACH (US), SEPA Direct Debit (EU). High return risk (NSF).
- **Economics**: Flat low fees (<$0.20 or basis points), no interchange cut.

Delayed / Vouchers

Offline cash vouchers generated online, paid at physical retail locations.

- **Boleto Bancário (Brazil)**: Barcode ticket paid at bank/lottery shops.
- **OXXO (Mexico)**: Cash voucher paid at convenience stores.
- **Konbini (Japan)**: Convenience store cash payment.
- **Latency**: Asynchronous confirmation (hours to days). Zero chargeback risk.

Digital Wallets

Container instruments managing credentials and checkout friction.

- **Pass-Through Wallets**: Apple Pay, Google Pay. Tokenizes underlying card (DPAN/FPAN).
- **Stored-Value Wallets**: PayPal, Alipay, WeChat Pay. Closed-loop balances with merchant ecosystems.
- **Conversion Boost**: 1-click biometric authentication lifts mobile checkout.

Crypto / Stablecoins

Public ledger rails providing programmable global settlement.

- **USDC / USDT**: Dollar-pegged assets on high-throughput L1/L2s (Solana, Base).
- **Advantages**: 24/7/365 settlement, sub-second finality, zero cross-border FX margins.
- **Handoffs**: Regulated on/off ramps handle local currency conversion.

SECTION 03 Rules, Compliance & Fraud Architecture

Regulation

- **Durbin Amendment (US)**: Caps debit interchange for banks >$10B to $0.21 + 0.05%. Exempts fintech credit and small bank debit.
- **IFR (Europe)**: Caps interchange at 0.20% (debit) and 0.30% (credit).
- **PSD2 / PSD3 & SCA**: Mandates multi-factor authentication on electronic European transactions.

Compliance & Data

- **PCI-DSS**: Security standard for storing/processing cardholder data. Tokenization (e.g. Stripe Elements) reduces audit scope to SAQ-A.
- **KYC / AML / OFAC**: Mandatory customer identification, beneficial ownership checks, and sanction screening.
- **Settlement Holding**: Rolling reserves held to protect against sudden chargeback spikes.

Fraud Typologies

- **Card-Not-Present (CNP)**: Identity theft and stolen card credentials testing.
- **Friendly Fraud**: Cardholder falsely claims unrecognized charge or non-delivery.
- **Account Takeover (ATO)**: Credential stuffing targeting stored wallet payment methods.
- **Card Testing**: Bot networks testing thousands of stolen PANs with micro-charges.

SECTION 04 Business Models: Funds Flow Comparison

Subscriptions & SaaS

Recurring Merchant-Initiated Transactions (MIT) following initial Customer-Initiated (CIT) setup.

- **Card Account Updater (CAU)**: Automated updating of expired card numbers across networks.
- **Smart Dunning**: Intelligent retries timed to consumer payday cycles to prevent involuntary churn.
- **Mandatory Auth Flags**: Stored credential indicators (credential-on-file) required by Visa/MC.

Marketplaces & Platforms

Tri-party funds flow: Buyer pays platform, platform takes take-rate, seller receives payout.

- **Merchant of Record (MoR)**: Platform handles all tax, compliance, and dispute liability (e.g. Gumroad).
- **PayFac / Split Payments**: Platform routes gross to sub-merchant accounts and claims commission (e.g. Shopify).
- **Money Transmission Licenses (MTL)**: Holding buyer funds requires escrow compliance or bank partnership.

B2B & Enterprise

High-value invoice matching, structured purchase orders, and Net 30/60 trade credit terms.

- **Level 2 & Level 3 Data**: Passing line-item invoice data, tax IDs, and freight codes cuts interchange by 0.50%–1.00%.
- **Virtual Cards**: Single-use dynamic PANs generated per PO with exact amount & supplier lock.
- **Reconciliation**: Automated matching against ERP systems (NetSuite, SAP).

SECTION 05

Key Takeaways: 10 Core Principles of Global Payments

1. Payments is Risk Pricing: Every basis point in fees reflects who bears fraud, chargeback, or settlement default risk.

2. Interchange Drives Consumer Credit: The 2% card fee isn't network profit—it funds points, cash back, and issuer credit lines.

3. Auth is a Promise, Settlement is Cash: An authorization holds funds; final transfer occurs hours or days later during batch clearing.

4. Asymmetric Underwriting: Issuers underwrite consumers; acquirers underwrite merchants against delivery failure.

5. Local Payment Methods Win Globally: Cards represent under 30% of e-commerce in LatAm and Asia; local rails (Pix, UPI, wallets) dominate conversion.

6. Tokenization Decouples Compliance: Storing network tokens instead of raw PANs shrinks merchant PCI scope while boosting authorization rates.

7. Chargeback Ceilings are Fatal: Crossing 1% dispute-to-transaction ratio triggers card brand fine programs (VDMP/VFMP) and processor termination.

8. MoR vs. PayFac Trade-Off: Being Merchant of Record takes on global sales tax and dispute liability; PayFac pushes liability to sellers.

9. Real-Time Rails Flatten Intermediaries: Instant A2A rails (FedNow, Pix) collapse multi-day settlement windows to seconds with negligible fees.

10. Conversion vs. Fraud Equilibrium: The highest payment revenue comes not from zero fraud, but from eliminating false positive declines on legitimate buyers.

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